Loumi TREMAS

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Professional Summary

PhD candidate in AI specializing in deep generative models, scalable ML infrastructure, and complex system simulation. Experienced in building high-performance world models (surrogates) and deploying neural networks on distributed GPU clusters. Proven track record of turning research prototypes into accelerated production systems ($10^6\times$ speedup) at STMicroelectronics.

Technical Skills

AI & Machine Learning: PyTorch, JAX, Generative AI (Diffusion, World Models), Reinforcement Learning (RL), Computer Vision.
Software & Infrastructure: Python, C, Distributed Computing (Ray), GPU Acceleration (CUDA), Linux, Azure.
Data & Engineering: Data Pipelines, Automated Benchmarking, Anomaly Detection, Root Cause Analysis, FDTD.
Languages: French (Native), English (C1)

Professional Experience

Teaching Assistant — Dec 2023 – Apr 2024
Savoie Mont-Blanc — Chambéry, France

- Supervised Master's students in Socio-Cognitive Ergonomics of Intelligent Systems, guiding AI-focused group projects and scalable ML practices.

AI Engineer — Aug 2022 – Feb 2023
STMicroelectronics — Crolles, France

- Built multimodal data pipelines and computer vision algorithms for automated anomaly detection in SEM images.
- Collaborated cross-functionally to harden research prototypes into reliable evaluation tools for production.

Education

PhD in AI & Physics — 2023 – 2026
University of Paris-Saclay & STMicroelectronics — Institut d'Optique Graduate School

- Engineered large-scale deep learning world models (surrogates) to simulate complex physical systems, achieving a $10^6\times$ inference acceleration.
- Designed training loops and custom generative models (Diffusion Schrödinger bridges) for inverse design optimization.
- Orchestrated distributed model training and profiling on GPU clusters for reliability and performance.

Master in Artificial Intelligence — 2022
Grenoble Ensimag — MSc in Informatics

- Specialized in Machine Learning Systems, Deep Learning, Numerical Methods, and Computer Vision.

Conferences and Articles

Selected Papers — **2026**
- Diffusion Schrödinger bridges for metasurface inverse design
- Enhanced posterior sampling via diffusion models for efficient metasurfaces inverse design

SPIE 2026, SPIE Photonics Europe — **Strasbourg, FRA**
- Heuristic-Initialized Gradient Descent for Inverse Design of Large-Scale Metasurfaces

SISPAD (Int. Conf. on Simulation of Semiconductor) — **2024 – 2025**
- Inverse Design of Optical Metasurface for CMOS Imagers: Multi-Objective Approach
- Transition to Solutions Based on Neural Networks for Flat Optical Devices

META, International Conference on Metamaterials — **2023 – 2026**
- *Self-enhanced generative metasurface inverse design* (Dublin, Ireland, 2026)
- *Impact of numerical simulation boundary-decomposition errors* (Toyama, Japan, 2024)

Interests & Hobbies

- **Endurance Sports:** Running and triathlon.
- **Automotive:** Finished **4th place** in the **4L Trophy** (1000+ teams), a 6,000 km humanitarian rally across the Moroccan desert.